

MSCA → Papers

Two Conference Papers from the CERTIFLIGHT Proposal

RobustUAVs.ai research programme · Roger Nick Anaedevha · 2026-10-01
Companion to the integration plan for Papers I and II

1 The decision

The CERTIFLIGHT MSCA proposal has four objectives (O1–O4) resting on four theorems (T1–T4) across five work packages. Two of them carry a self-contained technical result strong enough for a top venue *now*, on the preliminary work already in hand. Each becomes one paper.

- **Paper III** (NeurIPS) — from O3/T3/T4, WP4. *Certificates as Constraints*. The certified-constrained multi-agent controller: a formal robustness certificate used as a time-varying, pointwise CMDP constraint, with feasibility preserved during learning.
- **Paper IV** (USENIX Security) — from O1/O2, T1/T2, WP2/WP3. *When the Attacker Is Indistinguishable from the Weather*. The unified perturbation budget and cross-layer benchmark, reframed around the security observation that an attacker can disguise interference as a fault or a gust.

O4/WP5 (the assurance mapping) is *not* a third paper here: it is Paper II of the earlier batch (the SoK), which already absorbed it. T1’s tightness-gap result lives in Paper IV; T2’s interface-rate result is already in the parent manuscript and in Paper I.

2 Why these two venues

Paper	Venue	Why it fits
III	NeurIPS	The result is about the <i>quantifier</i> on a learned constraint — pointwise vs. in-expectation, under a constraint set that is itself a certified, adversarially moving object. A safe-RL / learning-theory contribution. The swarm is the setting, not the claim.
IV	USENIX Security	A threat-model correction (the attacker who dresses as the weather), an evaluation methodology with enforced provenance, and a released benchmark artifact. The shape USENIX rewards, and a venue where an evaluation substrate can become a community default.

3 Which MSCA theorem each paper carries

MSCA	What it is	Where it lands
T1	Heterogeneous-source composition with a quantified tightness gap	Paper IV , as the Makarov / VaR-aggregation characterisation of the union-bound slack.
T2	Interface theorem, discrete detection → continuous budget; γ as a supremum over an ODD	Parent manuscript (proved) and Paper I (the γ side); the budget side is Paper IV .
T3	Feasibility preservation for certified-constrained policies	Paper III , Propositions 1–2.
T4	Coverage-degradation bound for fleets with degraded agents	Paper III , Propositions 3–4.

4 How the four papers relate

Paper	Object, and its place in the chain $\theta \rightarrow \Delta \rightarrow \delta \rightarrow \text{MCR}$
Parent	Proves the chain for the time-delay class.
I (S&P)	Treats Δ and γ as <i>design variables</i> (authentication as credit and debit).
II (SoK)	Systematises the whole field by what its guarantees <i>quantify over</i> .
IV (USENIX)	Broadens the <i>source</i> of δ (fault, gust, disguised attack) and builds the benchmark that measures it.
III (NeurIPS)	Consumes the per-agent radius the chain produces, as a <i>fleet controller</i> constraint.

The four are non-overlapping in object: a certificate with fixed inputs (parent), a certificate with tunable inputs (I), the field’s certificates in general (II), a certificate against disguised sources (IV), and a controller that consumes certificates (III). Each cites the others; none repeats another’s result.

5 Readiness, honestly

Paper	Done	To do before submission
III	Propositions 1–4 proved and brute-force-checked (3000/3000); simulator and certificate exist	the CMDP wrapper, projection layer, degradation schedule, and the RL study
IV	Six-corpus ingest, certificate engine, residual-budget lemma, provenance flags, γ campaign	the fault/environment budget terms, the Makarov reduction (or its empirical fallback), the evasion construction, benchmark packaging and release

■ Two claims are not yet established and are flagged as such in the papers. In Paper IV, that the three-source composition reduces cleanly to a fixed-marginals-unknown-copula problem (Makarov) is a reduction to prove, with the sound union bound as the stated fallback. In Paper III, the empirical RL results do not yet exist; the three propositions are the contribution and already hold, so the paper degrades to a theory-plus-feasibility-layer result if MARL training is unstable at the target fleet size. Neither paper’s headline depends on an unproven claim.